

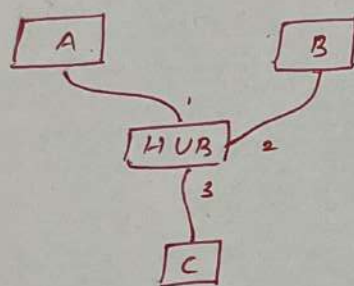
Day-7: Introduction of HUB:

HUB:

- It is used to connect multiple network device.
- It is a layer 1 device
- It is a broadcast device.

* Maximum 3-4 nic cards can be added to a PC to form network connection.

* But if we want more no. of devices then we have to use hub.



- When A wants to send a data to B, it sends a broadcast.
- But only B can open the message because destination mac will be given.
- C can receive the data but cannot open it, so C will drop the message.

DISADVANTAGE:

- * Since it is a broadcast device, network traffic is more, so network → slow down
- * Collision.

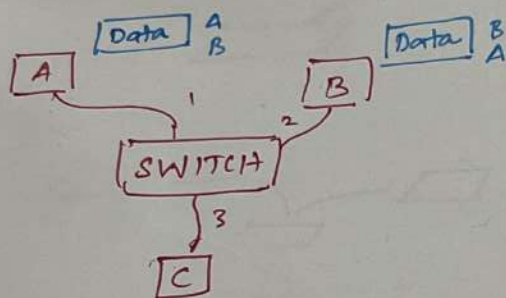
→ Because of the disadvantages in using hub, we can use switch to connect multiple devices.

SWITCH:

→ It is a layer 2 device
→ It is a broadcast device and used to connect multiple network device.

→ It contains MAC table

→ It act as broadcast + unicast



MAC TABLE	
PORT	SOURCE MAC
1	A
2	B

→ Address Learning.

When a data is sent to switch:

- ① Address Learning
- ② Check whether destination MAC is there already in table.

If it is already present

Unicast

It is not present

Broadcast

→ So switch is called "Intelligent Device" because it can decide whether to broadcast or unicast.

Different switches available:

- 1) Dlink
- 2) TPLink
- 3) Netgear

Ports available:

4/8/16/24/48

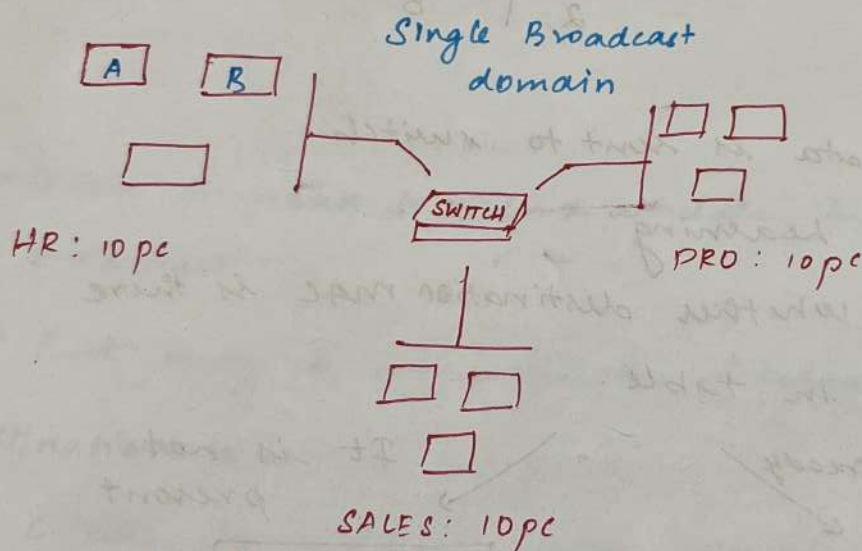
Speeds:

Ethernet 10 Mbps

Fast 100 Mbps

Giga 1000 Mbps

1 Gbps



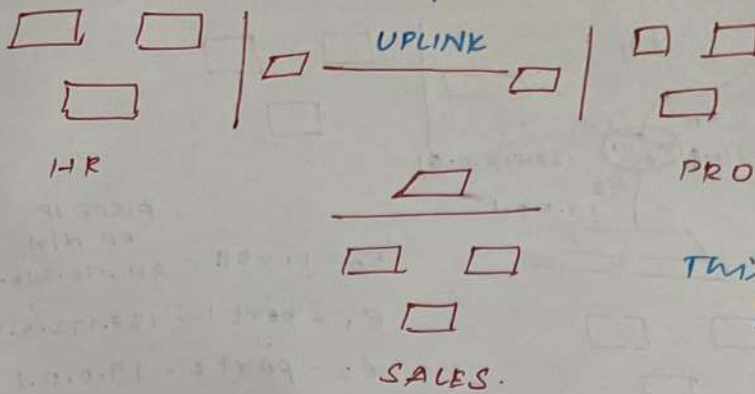
→ When 'A' pc wants to ping 'B', it will send ARP request:

S: 0A

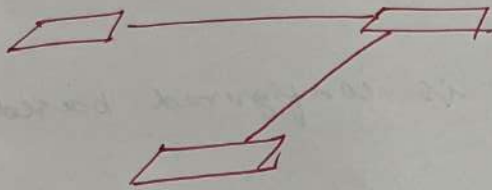
D: FFFF → Broadcast

So remaining 29 devices will get the ARP request. This is called Single Broadcast Domain. It cause heavy noise.

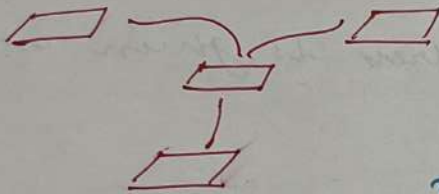
→ Connecting 2 Switches.



This has 2 broadcast domains



This is single broadcast domain



This is also single broadcast domain.

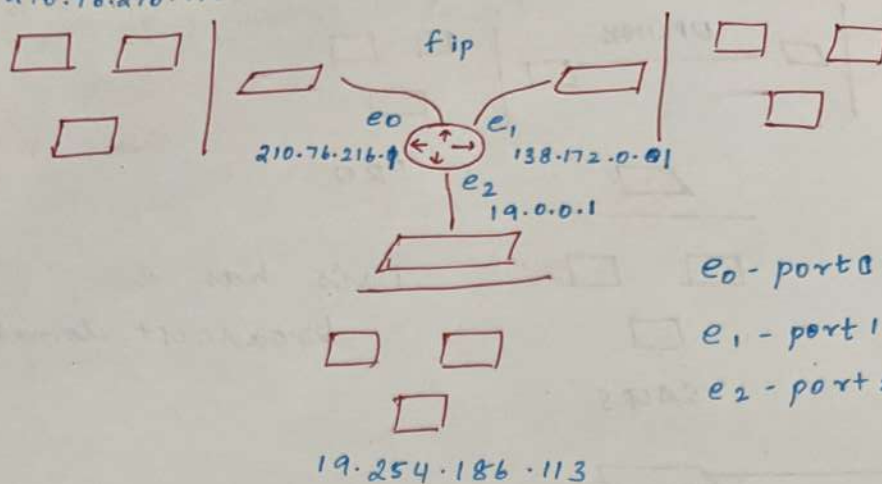
To avoid the single broadcast domain, we can use router

ROUTER:

- Used to break broadcast domain.
- Layer 3 device
- Unicast device
- It contains routing table.
- Routing table contains list of route
- It is used to connect different network.
- Unlike switch & hub, router is given IP configuration.

210.76.216.111

138.172.18.72



FIRST IP OF N/W
 e0 - port 0 - 210.76.216.1
 e1 - port 1 - 138.172.0.1
 e2 - port 2 - 19.0.0.1

* Each port in the router is configured based on the network.

* Each device has different network and the first IP of the network's IP address is given as the configuration of router.

ROUTING TABLE

FORMAT: Network ID, prefix, interface name.

Routing	
210.76.216.0	24 - e0
138.72.0.0	16 - e1
19.0.0.0	8 - e2

Once we assign ^{IP} address to the router automatically these values are stored in the routing table.

It is called as directly connected routes

Once a data enters the router, router will check whether there is a route for the destination IP.

Eg: Data.

S: 210.76.216.111

D: 19.254.186.113

→ The network ID of this data is already there in the routing table, so data will be unicasted to ez alone.